



# Customer Shopping Behavior Analysis

3,900 purchases

18 columns

Goal: uncover spending, segments, preferences, subscriptions



# Dataset Summary

## Scope

Rows: 3,900

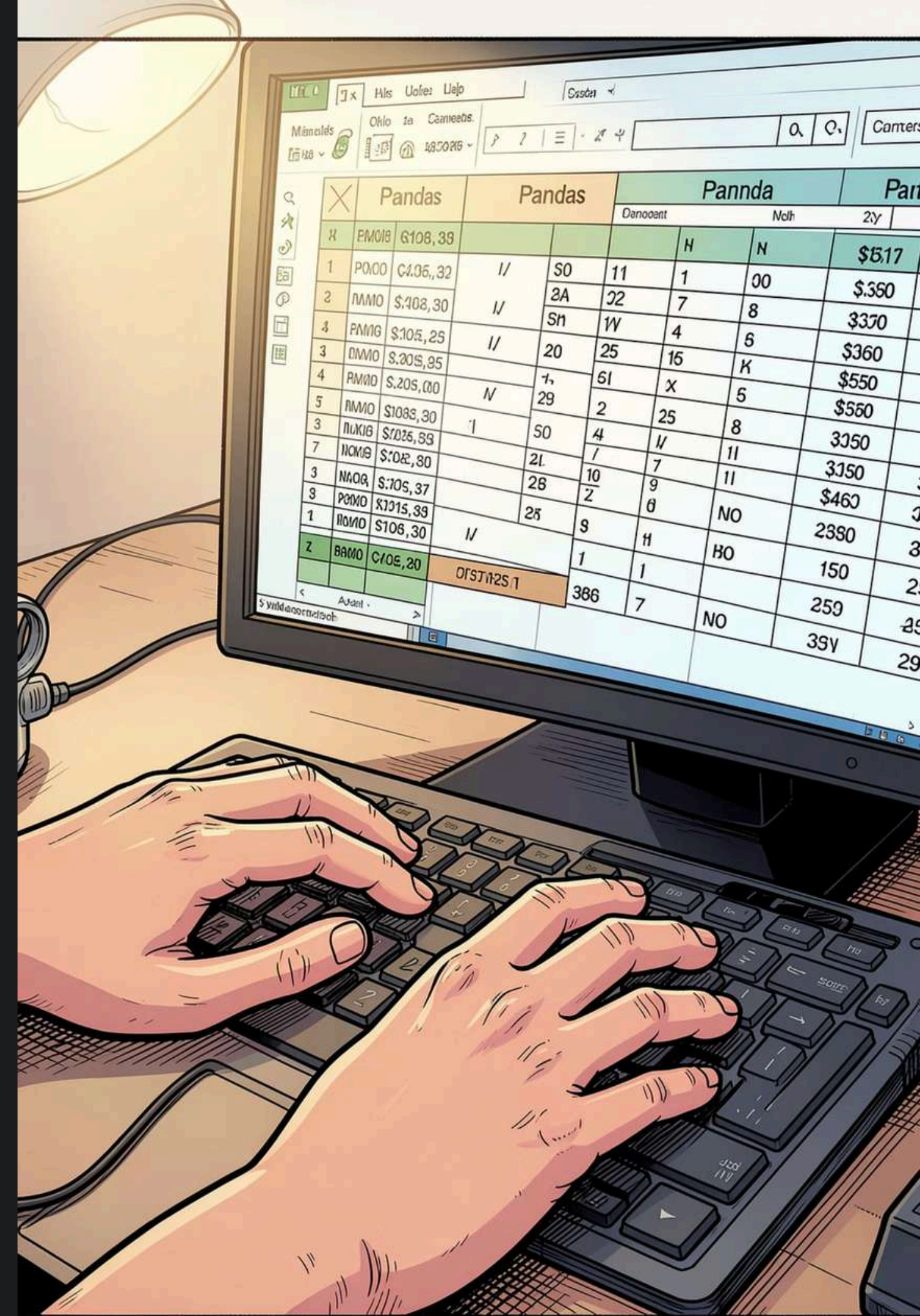
Columns: 18

## Key features

Age, Gender, Location, Item,  
Category, Amount, Season,  
Rating, Subscription, Shipping

## Missing

37 values in Review Rating





# Python — EDA & Preparation

01

## Load & Inspect

pandas: df.info(), df.describe()

02

## Impute Ratings

Median by product category for Review Rating

03

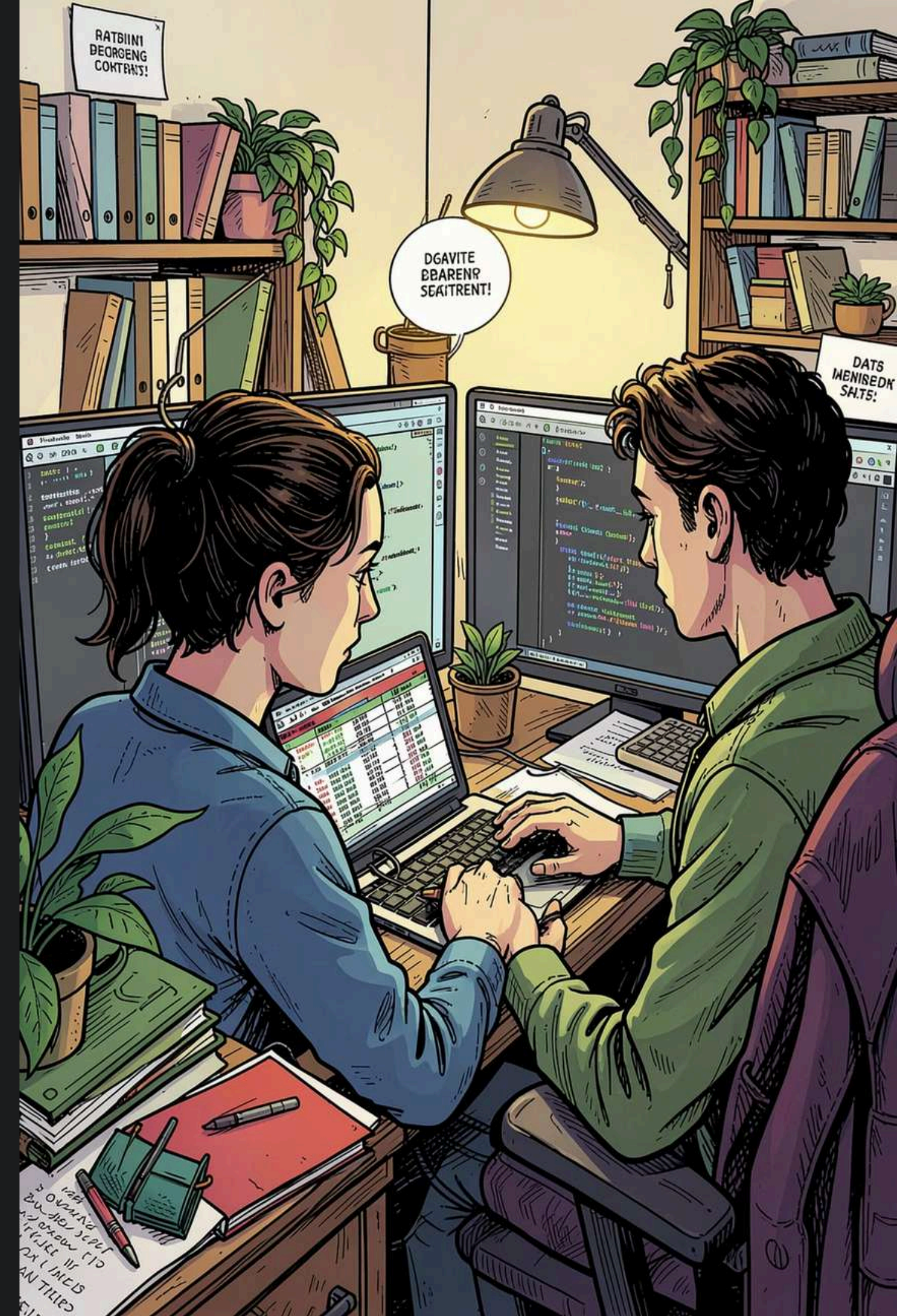
## Standardize & Engineer

snake\_case columns; age\_group; purchase\_frequency\_days

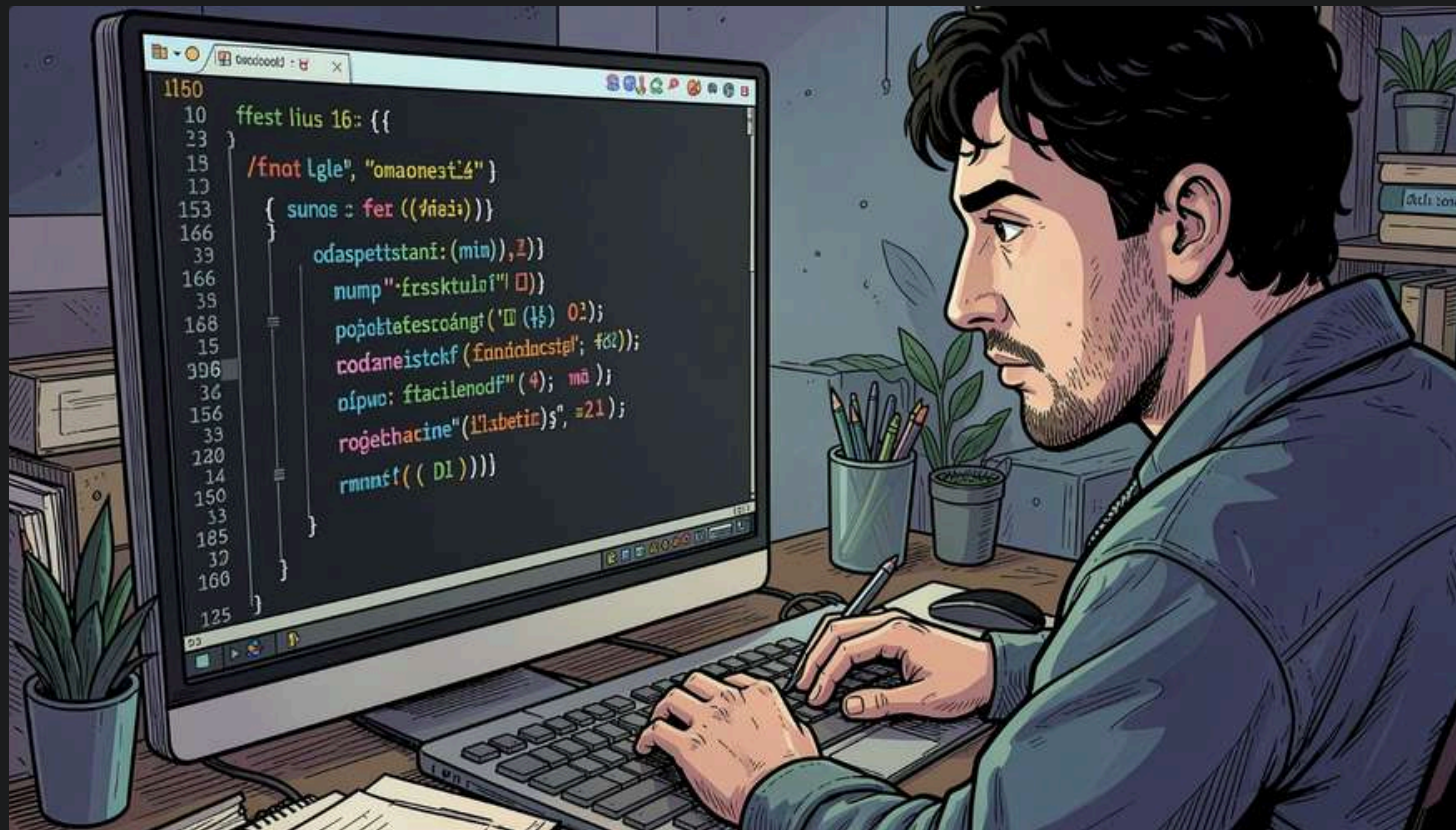
04

## DB Load

Export cleaned DataFrame to PostgreSQL







# SQL — Business Questions

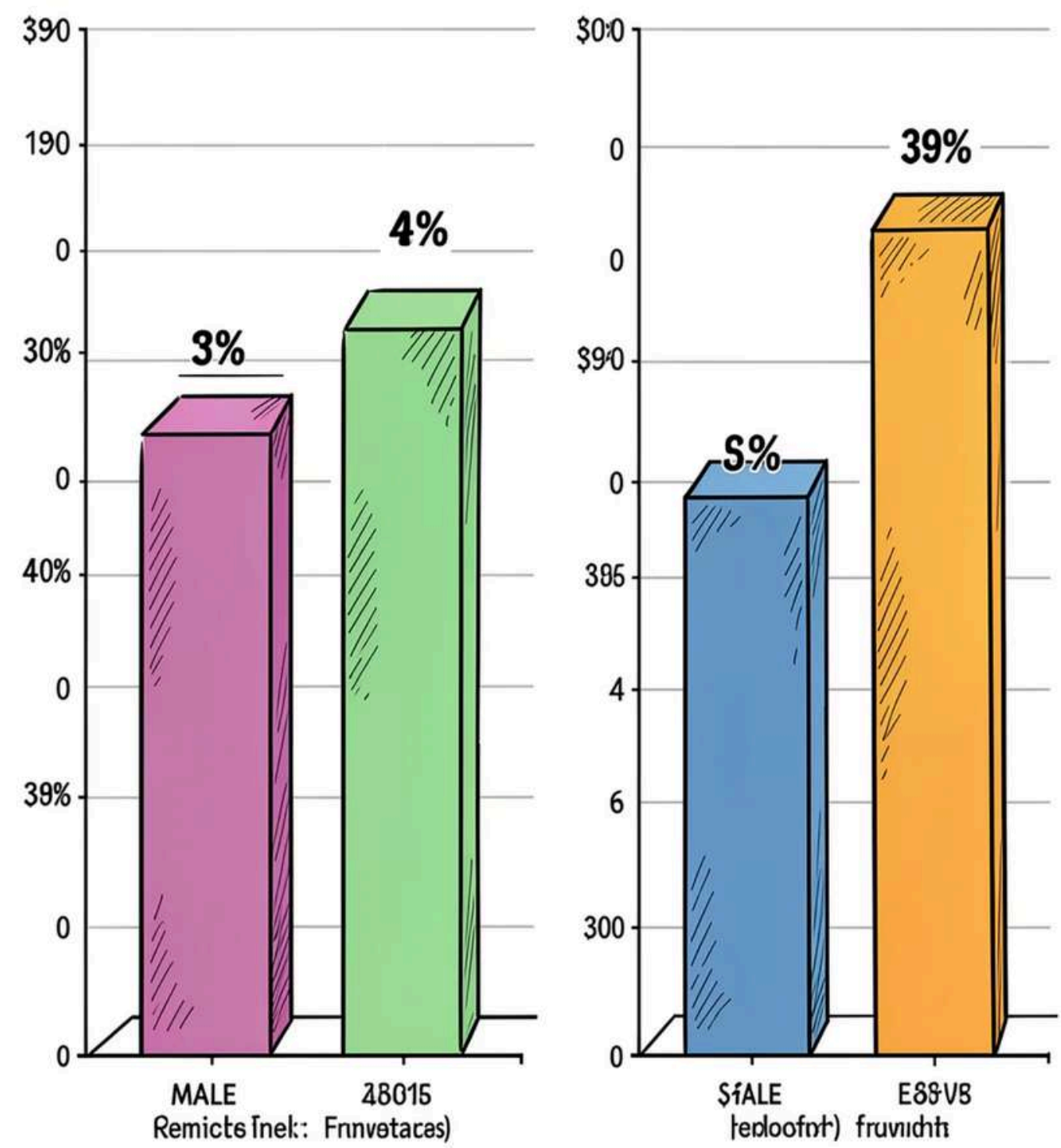
Revenue by gender • High-spending discount users • Top-rated products • Shipping comparison • Subscribers vs non-subscribers

📄 839 rows identified as discount users spending above average

# Revenue by Gender

|                         |
|-------------------------|
| Male                    |
| \$157,890 total revenue |
| Female                  |
| \$75,191 total revenue  |

MALES vs FEMALE REVENUE







# Discounts & Products



Hat  
50% discounted purchases



Sneakers  
49.66% discount rate



Coat  
49.07% discount rate



Sweater  
48.17% discount rate



Pants  
47.37% discount rate



# Top Products by Rating



Gloves

Avg rating 3.86



Sandals

3.84



Boots

3.82



Hat

3.80



Skirt

3.78

# Power BI Dashboard Highlights

## KPIs

3.9K customers · \$59.76 avg purchase · 3.75 avg rating

## Subscription Split

Yes 27% · No 73%

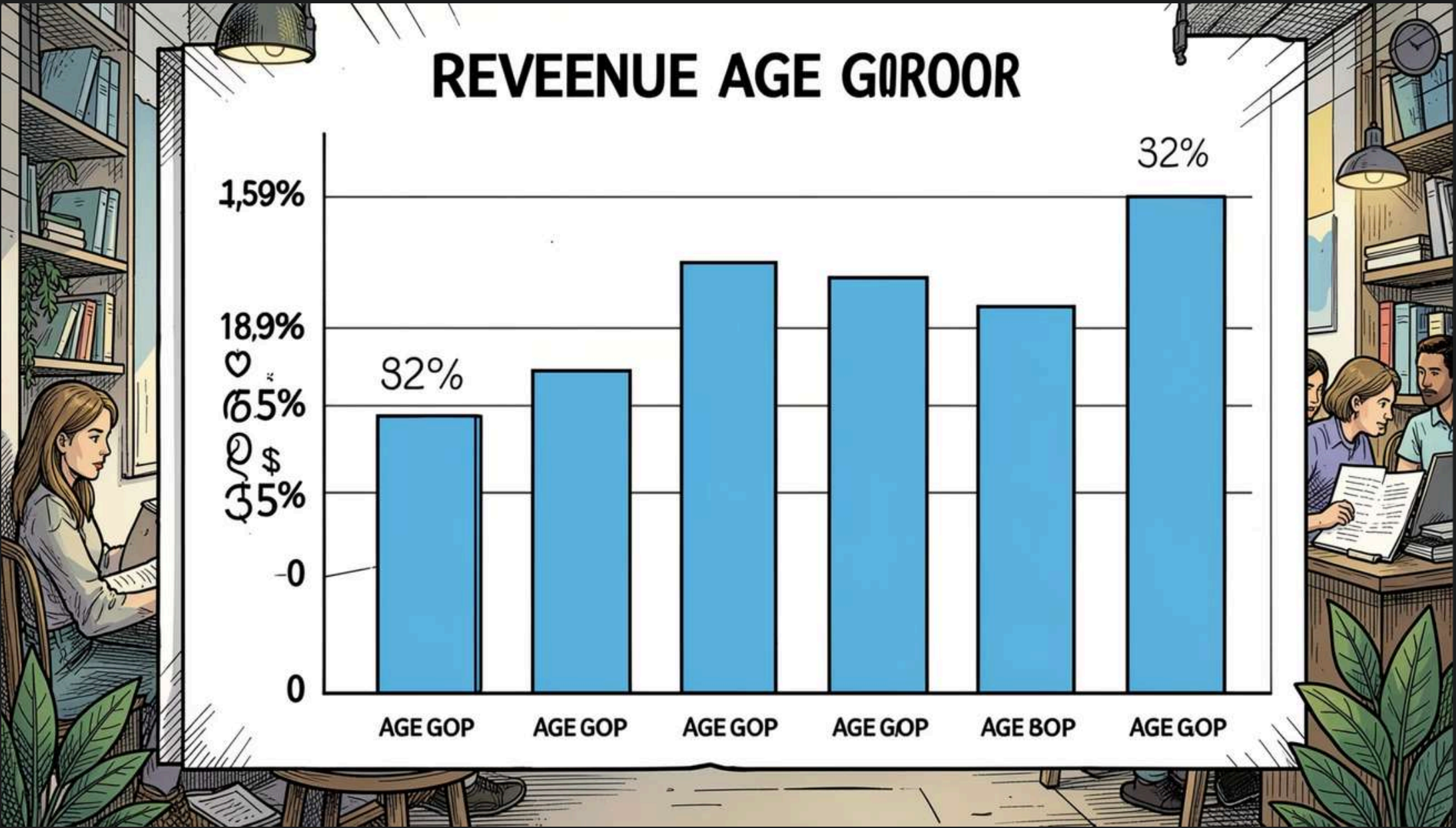
## Revenue by Category

Clothing ~100K · Accessories ~60K · Footwear ~30K · Outerwear ~10K





# Customer Segments & Revenue



## Segments

Loyal 3,116 · Returning 701 · New 83

## Revenue by Age

Young Adult 62,143 · Middle-aged 59,197 ·  
Adult 55,978 · Senior 55,763

## Shipping

Standard avg \$58.46 · Express avg \$60.48



# Business Recommendations

## Boost Subscriptions

Promote exclusive subscriber benefits

## Loyalty Programs

Reward repeat buyers to increase loyal segment

## Review Discounts

Balance sales lift with margin control

## Product Positioning

Highlight top-rated & best-selling items

## Targeted Marketing

Focus on high-revenue age groups & express-shipping users